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Author- Swbaig24

PRODUCT REQUIREMENTS DOCUMENT (PRD)

AI-Powered Multi-Channel Customer Support Automation Platform

****Version:**** 1.0

****Date:**** [Today's Date]

****Author:**** [Your Name]

****Status:**** In Development

1. EXECUTIVE SUMMARY

1.1 Product Vision

A generic, industry-agnostic AI automation platform that provides 24/7 customer support across multiple channels (Website Chat, Email, WhatsApp, Voice) with customizable industry-specific knowledge bases.

1.2 Problem Statement

Small to medium businesses across all industries lack affordable, intelligent customer support automation that can handle common queries without human intervention.

1.3 Solution

A no-code/low-code AI automation ecosystem that:

- Deploys in under 1 hour
- Costs near-zero to start
- Works across 4 channels (Chat, Email, WhatsApp, Voice)
- Adapts to any industry via customizable knowledge bases

2. PRODUCT SCOPE

2.1 In Scope (MVP - Phase 1)

****Website Chatbot****

- Real-time conversational AI
- Industry-specific Q&A
- Handoff to human agent (future)
- Multi-language support (future)

****Email Auto-Responder****

- Automatic email classification
- Smart response generation
- Escalation rules

****WhatsApp Business Integration****

- Text-based support

- Rich media responses
- Business hours automation

✓ ****AI Voice Agent**** (Phase 2)

- Inbound call handling
- Natural voice conversations
- Appointment scheduling

2.2 Out of Scope (For Now)

- ✗ Custom mobile app
- ✗ Advanced analytics dashboard
- ✗ Multi-tenant SaaS architecture
- ✗ Payment processing integration

3. USER PERSONAS

3.1 Primary User: Business Owner

- ****Goal:**** Reduce customer support costs by 60%
- ****Pain Point:**** Can't afford 24/7 human support staff
- ****Success Metric:**** 80% of queries resolved without human intervention

3.2 End User: Customer

- ****Goal:**** Get instant answers to common questions
- ****Pain Point:**** Long wait times for email/phone support
- ****Success Metric:**** Response time < 5 seconds

4. FUNCTIONAL REQUIREMENTS

FR-1: Multi-Channel Support

| ID | Requirement | Priority | Status |

|----|-----|-----|-----|

| FR-1.1 | Deploy chatbot on any website via embed code | High | ✓ Done |

| FR-1.2 | Send/receive WhatsApp messages | High | 🔄 In Progress |

| FR-1.3 | Auto-respond to emails | High | ● Not Started |

| FR-1.4 | Handle voice calls | Medium | Not Started |

FR-2: AI Intelligence

| ID | Requirement | Priority | Status |

|----|-----|-----|-----|

| FR-2.1 | Answer industry-specific questions | High | ✓ Done |

| FR-2.2 | Handle out-of-scope queries gracefully | High | ✓ Done |

| FR-2.3 | Maintain conversation context | Medium | 🔄 In Progress |

| FR-2.4 | Learn from past interactions | Low | ● Not Started |

FR-3: Knowledge Management

| ID | Requirement | Priority | Status |

|----|-----|-----|-----|

| FR-3.1 | Upload custom knowledge base (text/PDF) | High |  Done |

| FR-3.2 | Update knowledge base without downtime | High | Not Started |

| FR-3.3 | Multi-industry template library | Medium | Not Started |

5. TECHNICAL ARCHITECTURE

5.1 Technology Stack

| Component | Tool | Purpose | Cost |

|-----|-----|-----|-----|

| ****AI Chat Engine**** | Voiceflow | Conversational AI logic | Free tier |

| ****Workflow Automation**** | Make.com | Connect channels & triggers | Free tier |

| ****Knowledge Base**** | Voiceflow KB | Store industry-specific data | Free tier |

| ****Website Hosting**** | Carrd.co | Landing page & demo | Free tier |

| ****Version Control**** | GitHub | Documentation & code | Free |

| ****WhatsApp API**** | Meta Business API | WhatsApp integration | Free (conversation-based pricing) |

| ****Voice Agent**** | Vapi.ai / Bland.ai | Phone call automation | Free tier |

5.2 System Flow

Customer → [Channel: Web/Email/WhatsApp/Voice]

→ Make.com (Webhook/Trigger)

→ Voiceflow API (AI Processing)

→ Knowledge Base Lookup

→ Response Generation

→ Make.com (Routing)

→ Customer

6. SUCCESS METRICS (KPIs)

| Metric | Target | Measurement |

|-----|-----|-----|

| ****Response Time**** | < 5 seconds | Voiceflow analytics |

| ****Resolution Rate**** | > 80% without human | Conversation logs |

| ****Customer Satisfaction**** | > 4/5 stars | Post-chat survey |

| ****Cost Savings**** | 60% vs human agent | Monthly cost comparison |

| ****Uptime**** | 99.5% | Make.com/Voiceflow status |

7. ROADMAP

Phase 1: MVP (Current - Weeks 1-4)

-  Website chatbot (DONE)
-  WhatsApp integration (IN PROGRESS)
-  Email auto-responder
-  Basic documentation

Phase 2: Scale (Weeks 5-8)

-  Voice agent integration
- Multi-industry templates
-  Analytics dashboard
-  Client onboarding system

Phase 3: Monetization (Weeks 9-12)

-  Pricing tiers
-  White-label options
- API access for developers
-  Partner program

8. RISKS & MITIGATION

Risk	Impact	Mitigation
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API rate limits on free tiers	Medium	Upgrade to paid tiers as revenue grows
WhatsApp API approval delays	High	Start approval process early
AI hallucinations (wrong answers)	High	Strict knowledge base boundaries, human review
Platform dependency (Voiceflow/Make)	Medium	Document architecture for future migration

9. COMPETITIVE ANALYSIS

Competitor	Weakness	Our Advantage
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Intercom	Expensive (\$74+/mo)	Near-zero cost to start
Zendesk AI	Complex setup	1-hour deployment
Custom chatbots	Require coding	No-code, generic template

10. APPENDICES

10.1 Glossary

- **KB**: Knowledge Base
- **MVP**: Minimum Viable Product
- **API**: Application Programming Interface

10.2 References

- Voiceflow Documentation: <https://docs.voiceflow.com>
- Make.com Documentation: <https://www.make.com/en/help>
- WhatsApp Business API: <https://business.whatsapp.com>